

help you get rid of the bulky chargers that come with laptops and other devices.

This power electronics solution is based on gallium nitride (as opposed to silicon)—a semiconductor that wastes far less power than the silicon components used in conventional devices.

"The current solution, which is based on silicon, has reached its limit in high-voltage power conversion. We can't eke out any more efficiency out of this technology. The time is now to do something different and to impact the 10 per cent of wasted energy that occurs in power conversion," explained Umesh Mishra, CEO and co-founder, Transphorm.

Laser to kill mosquitoes

Graphics may now prove useful in powering lasers that can zap live malaria-bearing mosquitoes out of the sky. Intellectual Ventures has come up with a laser fence that uses an Nvidia graphics chip to process the data coming in from a camera to identify a mosquito in real time and then activate the laser to zap it.

Earlier, DDT was sprayed as an effective solution to kill mosquitoes

but it was banned due to health concerns. The firm will employ the computing process of Nvidia's chips that use multiple cores to get through huge computing problems, in guiding lasers to identify and shoot down female mosquitoes.

"You have to recognise that it's a mosquito, versus all of the other bugs that might be flying around, recognise that it's a female mosquito, and then shoot it down. That recognition of trying to find that mosquito while it's flying around is done on the GPU," said Dan Vivoli, senior vice president of marketing at Nvidia.

Contactless hybrid finger scanner

NEC Corporation claims to have launched the world's first scanner that simultaneously acquires fingerprint and finger vein characteristics without requiring physical contact. The HS100-10 scanner's simultaneous acquisition



NEC HS100-10 fingerprint scanner

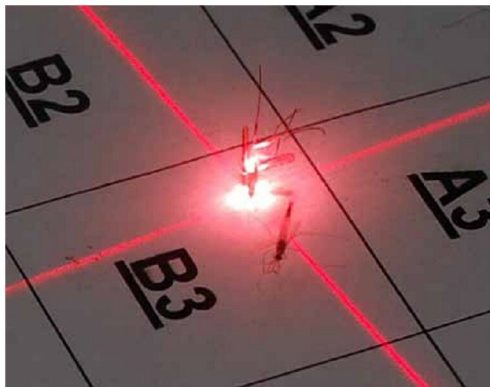
of both fingerprint and finger vein information makes it difficult for impostors to deceive the authentication system. Its contactless sensor is robust against the characteristics of fingers that are exceedingly dry or moist with exceptional accuracy, unlike conventional contact-based fingerprint authentication devices.

Furthermore, the USB2.0 interface allows users to connect the scanner to a PC in order to log on to its operating system or unlock its screensaver in both a standalone environment and an ActiveDirectory environment. The scanner is also compatible with major access management software. The combination of support for ID and access management with biometric data significantly reduces user concerns regarding the proliferation of user IDs and passwords.

Self-repairing chip

Researchers at CRISP consortium have conceptualised a chip that allows for 100 per cent functionality even with faulty components. The chip is basically a self-repairing multicore processor that allows on-chip components to keep on shrinking while combating concerns over accelerated degradation.

The consortium developed new concepts for run-time resource management to attain the goal of self-repair. While in operation, the chip tests cores and connections, and a resource



Intellectual Ventures' mosquito killer laser